

UNIT-4

Asymmetric Ciphers

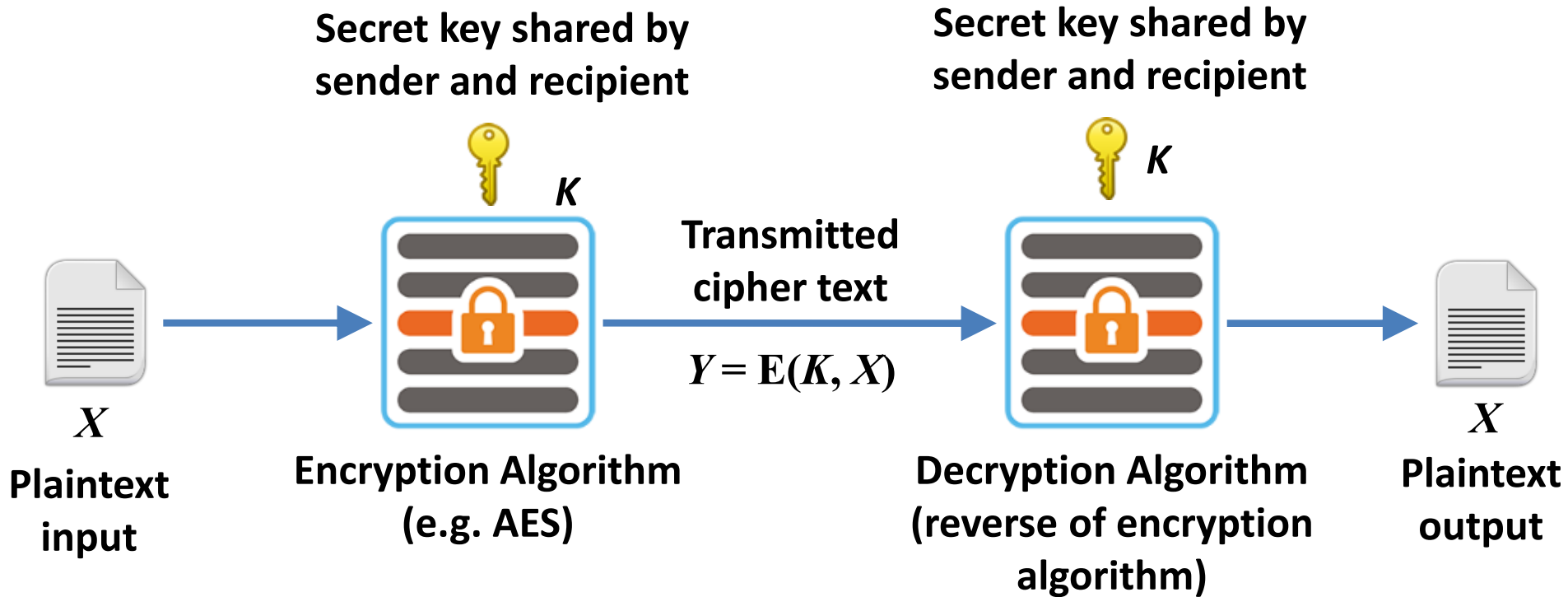


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Outline

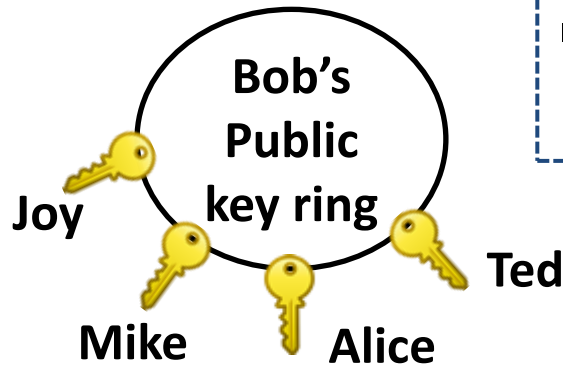
- Public Key Cryptosystems with Applications
- Requirements and Cryptanalysis
- RSA algorithm
- RSA computational aspects and security
- Diffie-Hillman Key Exchange algorithm
- Man-in-Middle attack

Symmetric key Encryption



Asymmetric key Encryption with Public Key

- The entire encrypted message serves as a **confidentiality**.



PU_a

Alice's public
key

PR_a

Alice's private
key



X



Transmitted
cipher text

$$Y = E(PU_a, X)$$



X



Plaintext
input

Encryption Algorithm
(e.g. RSA)

Decryption Algorithm

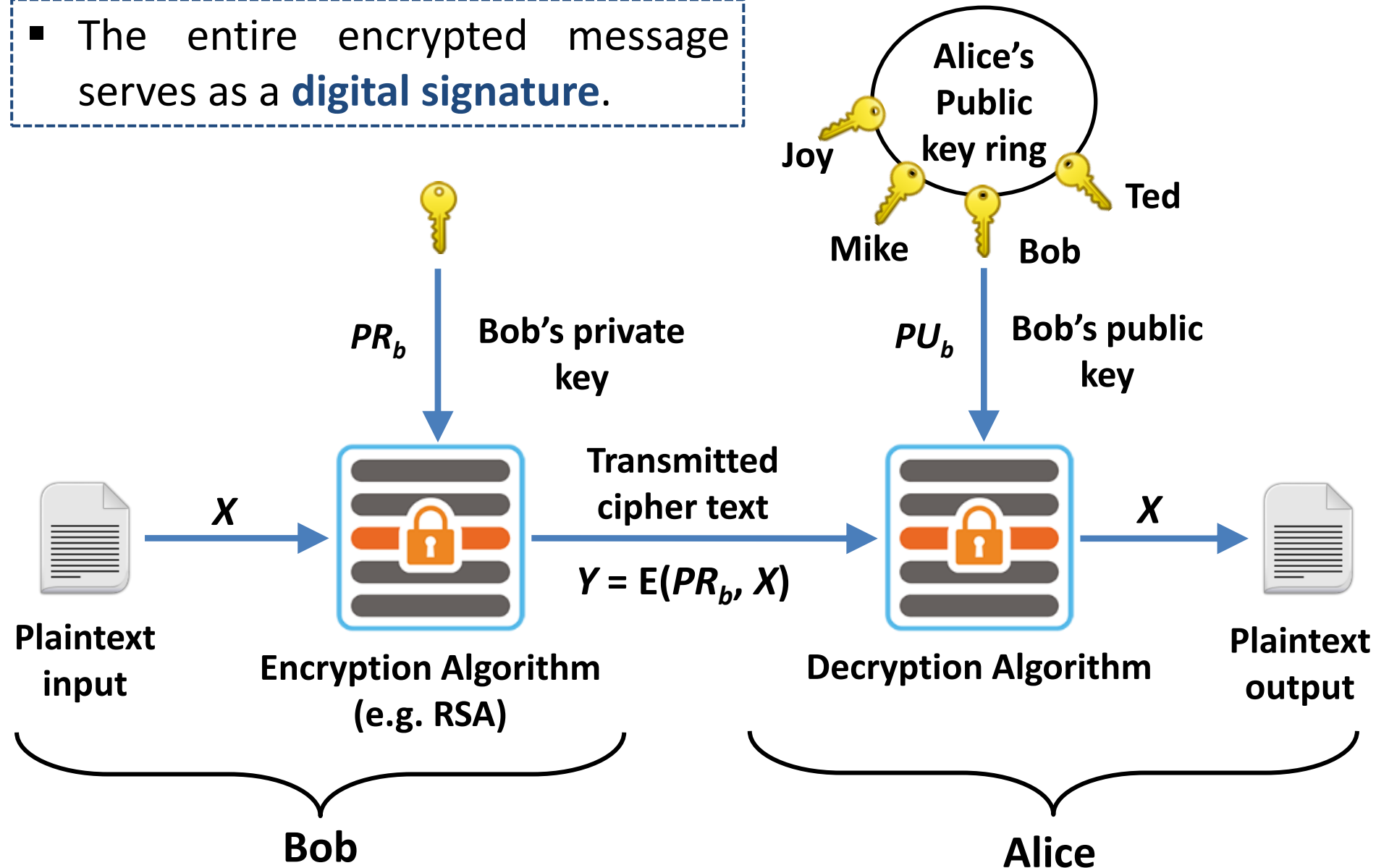
Plaintext
output

Bob

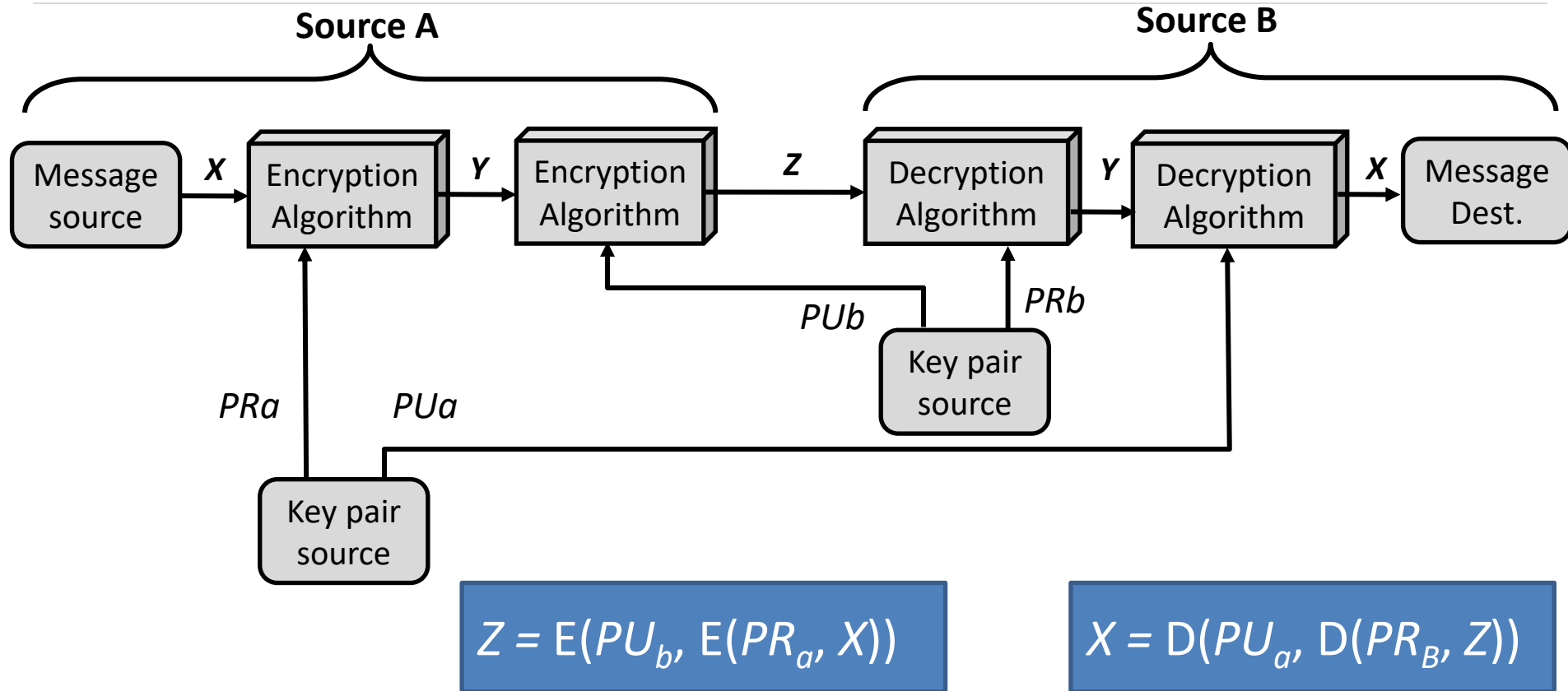
Alice

Asymmetric key Encryption with Private Key

- The entire encrypted message serves as a **digital signature**.



Authentication and Confidentiality



Applications for Public-Key Cryptosystems

- **Encryption/decryption:** The sender encrypts a message with the recipient's public key.
- **Digital signature:** The sender “signs” a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.
- **Key exchange:** Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

RSA Algorithm

- **RSA** is a block cipher in which the Plaintext and Ciphertext are represented as integers between **0** and **$n-1$** for some n .
- Large messages can be broken up into a number of blocks.
- Each block would then be represented by an integer.

Step-1: Generate Public key and Private key

Step-2: Encrypt message using Public key

Step-3: Decrypt message using Private key

Step-1: Generate Public key and Private key

- Select two large prime numbers: **p** and **q**
- Calculate modulus : **$n = p * q$**
- Calculate Euler's totient function : **$\phi(n) = (p-1) * (q-1)$**
- Select **e** such that **e** is **relatively prime** to **$\phi(n)$** and **$1 < e < \phi(n)$**

Two numbers are relatively prime if they have no common factors other than 1.

- Determine **d** such that **$d * e \equiv 1 \pmod{\phi(n)}$**
- Publickey : **$PU = \{ e, n \}$**
- Privatekey : **$PR = \{ d, n \}$**

Step-1: Generate Public key and Private key

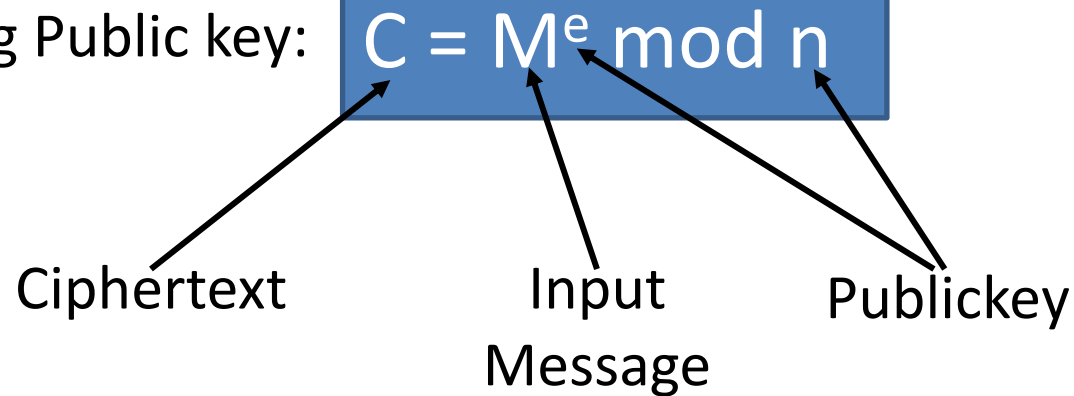
- Select two large prime numbers: $p = 3$ and $q = 11$
- Calculate modulus : $n = p * q, n = 33$
- Calculate Euler's totient function : $\phi(n) = (p-1) * (q-1)$
$$\phi(n) = (3 - 1) * (11 - 1) = 20$$
- Select e such that e is relatively prime to $\phi(n)$ and $1 < e < \phi(n)$
- We have several choices for $e : 7, 11, 13, 17, 19$ Let's take $e = 7$
- Determine d such that $d * e \equiv 1 \pmod{\phi(n)}$
- $? * 7 \equiv 1 \pmod{20}, 3 * 7 \equiv 1 \pmod{20}$

- Public key : $PU = \{ e, n \}, PU = \{ 7, 33 \}$
- Private key : $PR = \{ d, n \}, PR = \{ 3, 33 \}$

- This is equivalent to finding d which satisfies $de = 1 + j.\phi(n)$ where j is any integer.
- We can rewrite this as $d = (1 + j. \phi(n)) / e$

Step-2 : Encrypt Message

- Encryption Using Public key:



$PU = \{ e, n \}, PU = \{ 7, 33 \}$

For message $M = 14$

$$C = 14^7 \bmod 33$$

$$C = [(14^1 \bmod 33) \times (14^2 \bmod 33) \times (14^4 \bmod 33)] \bmod 33$$

$$C = (14 \times 31 \times 4) \bmod 33 = 1736 \bmod 33$$

$$C = 20$$

Step-3 : Decrypt Message

- Encryption Using Public key:

$$M = C^d \bmod n$$

Plaintext
Message

Cipher
Message

Privatekey

$$PR = \{ d, n \}, PR = \{ 3, 33 \}$$

For Ciphertext $C = 20$

$$M = 20^3 \bmod 33$$

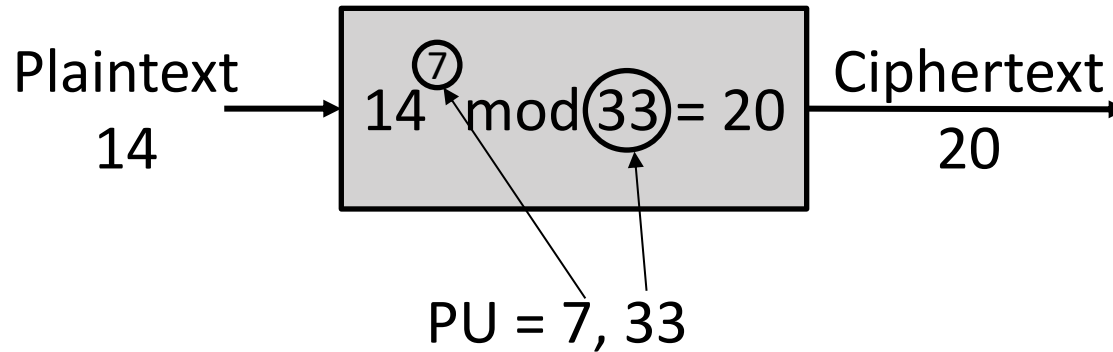
$$M = [(20^1 \bmod 33) \times (20^2 \bmod 33)] \bmod 33$$

$$M = (20 \times 4) \bmod 33 = 80 \bmod 33$$

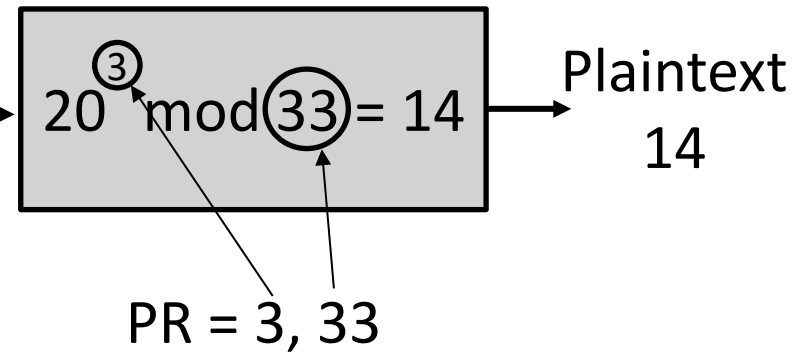
$$M = 14$$

Example RSA Algorithm

Encryption



Decryption



RSA Example

- Find n , $\phi(n)$, e , d for $p=7$ and $q=19$ then demonstrate encryption and decryption for $M=6$

$$n = p * q = 7 * 19 = 133$$

$$\phi(n) = (p - 1) * (q - 1) = 108$$

Finding e relatively prime to 108

$$e = 2 \Rightarrow \text{GCD}(2, 108) = 2 \text{ (no)}$$

$$e = 3 \Rightarrow \text{GCD}(3, 108) = 3 \text{ (no)}$$

$$e = 5 \Rightarrow \text{GCD}(5, 108) = 1 \text{ (Yes)}$$

- Finding d such that $(d * e) \bmod \phi(n) = 1$
- We can rewrite this as $d = (1 + j * \phi(n)) / e$
 - $j = 0 \Rightarrow d = 1 / 5 = 0.2 \leftarrow \text{integer? (no)}$
 - $j = 1 \Rightarrow d = 109 / 5 = 21.8 \leftarrow \text{integer? (no)}$
 - $j = 2 \Rightarrow d = 217 / 5 = 43.4 \leftarrow \text{integer? (no)}$
 - $j = 3 \Rightarrow d = 325 / 5 = 65 \text{ integer? (yes)}$

Public key :

$$\text{PU} = \{ e, n \} = \{ 5, 133 \}$$

Private key :

$$\text{PR} = \{ d, n \} = \{ 65, 133 \}$$

RSA Example – cont...

- Encryption:

$$C = M^e \bmod n$$

$$PU = \{ e, n \}, PU = \{ 5, 133 \}$$

For message $M = 6$

$$C = 6^5 \bmod 133$$

$$C = 7776 \bmod 133$$

$$C = 62$$

- Decryption:

$$M = C^d \bmod n$$

$$PR = \{ d, n \}, PR = \{ 65, 133 \}$$

For $C = 62$

$$M = 62^{65} \bmod 133$$

$$M = 2666 \bmod 133$$

$$M = 6$$

RSA Example

- P and Q are two prime numbers. $P=7$, and $Q=17$. Take public key $E=5$. If plain text value is 10, then what will be cipher text value according to RSA algorithm?
- $n = 119$
- $\phi(n) = 96$
- $e = 5$
- $d = 77$
- $PU = \{ 5, 119 \}$
- $PR = \{77, 119\}$
- $C = 10^5 \bmod 119 \Rightarrow C = 40$

Diffie-Hellman key Exchange

- The purpose of the **Diffie-Hellman** algorithm is to enable two users to securely exchange a key that can be used for subsequent encryption of message.
- This algorithm depends for its effectiveness on the difficulty of computing **discrete logarithms**.

Primitive root

- Let p be a prime number
- Then a is a primitive root for p , if the powers of a modulo p generates all integers from 1 to $p - 1$ in some permutation.

$$a \bmod p, a^2 \bmod p, \dots, a^{p-1} \bmod p$$

- Example: $p = 7$ then primitive root is 3 because powers of 3 mod 7 generates all the integers from 1 to 6

$$3^1 = 3 \equiv 3 \pmod{7}$$

$$3^2 = 9 \equiv 2 \pmod{7}$$

$$3^3 = 27 \equiv 6 \pmod{7}$$

$$3^4 = 81 \equiv 4 \pmod{7}$$

$$3^5 = 243 \equiv 5 \pmod{7}$$

$$3^6 = 729 \equiv 1 \pmod{7}$$

Discrete Logarithm

- For any integer b and a primitive root a of prime number p , we can find a unique exponent i such that

$$b = a^i \pmod{p} \text{ where } 0 \leq i \leq (p - 1)$$

- The exponent i is referred as the discrete logarithm of b for the base a , mod p . It expressed as below.

$$bdlog_{a,p}(b)$$

Diffie-Hellman Key Exchange – Cont...

- User A and User B agree on two large prime numbers q and α .
User A and User B can use insecure channel to agree on them.
- User A selects a random integer $X_A < q$ and calculates Y_A

Diffie-Hellman Key Exchange – Cont...

Global Public Elements

q prime number

α $\alpha < q$ and α is primitive root of q

User A Key Generation

Select private X_A $X_A < q$

Calculate public Y_A $Y_A = \alpha^{X_A} \bmod q$

User B Key Generation

Select private X_B $X_B < q$

Calculate public Y_B $Y_B = \alpha^{X_B} \bmod q$

Diffie-Hellman Key Exchange – Cont...

User A Key Generation

Select private X_A $X_A < q$

Calculate public Y_A $Y_A = \alpha^{X_A} \bmod q$

User B Key Generation

Select private X_B $X_B < q$

Calculate public Y_B $Y_B = \alpha^{X_B} \bmod q$

Calculation of Secret Key by User A

$K =$

Calculation of Secret Key by User b

$K =$

Diffie-Hellman Key Exchange – Cont...

User A Key Generation

Private X_A $X_A < q$,

Public Y_A

$$Y_A = \alpha^{X_A} \bmod q$$

User B Key Generation

Private X_B $X_B < q$,

Public Y_B

$$Y_B = \alpha^{X_B} \bmod q$$

Secret Key by User A : $K = (Y_B)^{X_A} \bmod q$

Secret Key by User B : $K = (Y_A)^{X_B} \bmod q$

$$K = (Y_B)^{X_A} \bmod q$$

$$K = (\alpha^{X_B} \bmod q)^{X_A} \bmod q$$

$$K = (\alpha^{X_B})^{X_A} \bmod q$$

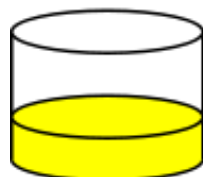
$$K = \alpha^{X_B X_A} \bmod q$$

$$K = (\alpha^{X_A})^{X_B} \bmod q$$

$$K = (\alpha^{X_A} \bmod q)^{X_B} \bmod q$$

$$K = (Y_A)^{X_B} \bmod q$$

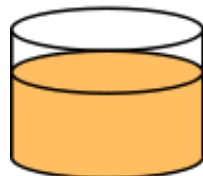
Alice



+



=



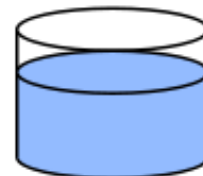
Bob



+



=

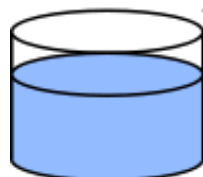


Common paint

Secret colours

Public transport

(assume
that mixture separation
is expensive)



+



=



+



=



Secret colours

Common secret

Diffie-Hellman Key Exchange Example

- Alice and Bob agree on a prime number $q = 23$
- $\alpha = 5$ as primitive root of q
- Alice selects a private integer $X_A = 6$
- Alice computes $Y_A = \alpha^{X_A} \bmod q \Rightarrow Y_A = 5^6 \bmod 23 = 8$
- Bob selects a private integer $X_B = 15$
- Bob computes $Y_B = \alpha^{X_B} \bmod q \Rightarrow Y_B = 5^{15} \bmod 23 = 19$
- Alice sends Y_A to Bob and Bob sends Y_B to Alice
- Alice computes key $K = (Y_B)^{X_A} \bmod q \Rightarrow K = (19)^6 \bmod 23$
- $K = 2$
- Bob computes key $K = (Y_A)^{X_B} \bmod q \Rightarrow K = (8)^{15} \bmod 23$
- $K = 2$

Diffie-Hellman Key Exchange Illustration

Alice

Bob

Alice and Bob share a prime number q and an integer α , such that $\alpha < q$ and α is a primitive root of q

Alice and Bob share a prime number q and an integer α , such that $\alpha < q$ and α is a primitive root of q

Alice generates a private key X_A such that $X_A < q$

Bob generates a private key X_B such that $X_B < q$

Alice calculates a public key $Y_A = \alpha^{X_A} \bmod q$

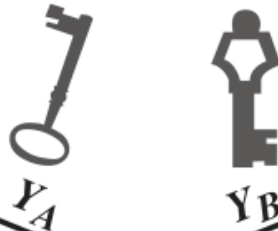
Bob calculates a public key $Y_B = \alpha^{X_B} \bmod q$

Alice receives Bob's public key Y_B in plaintext

Bob receives Alice's public key Y_A in plaintext

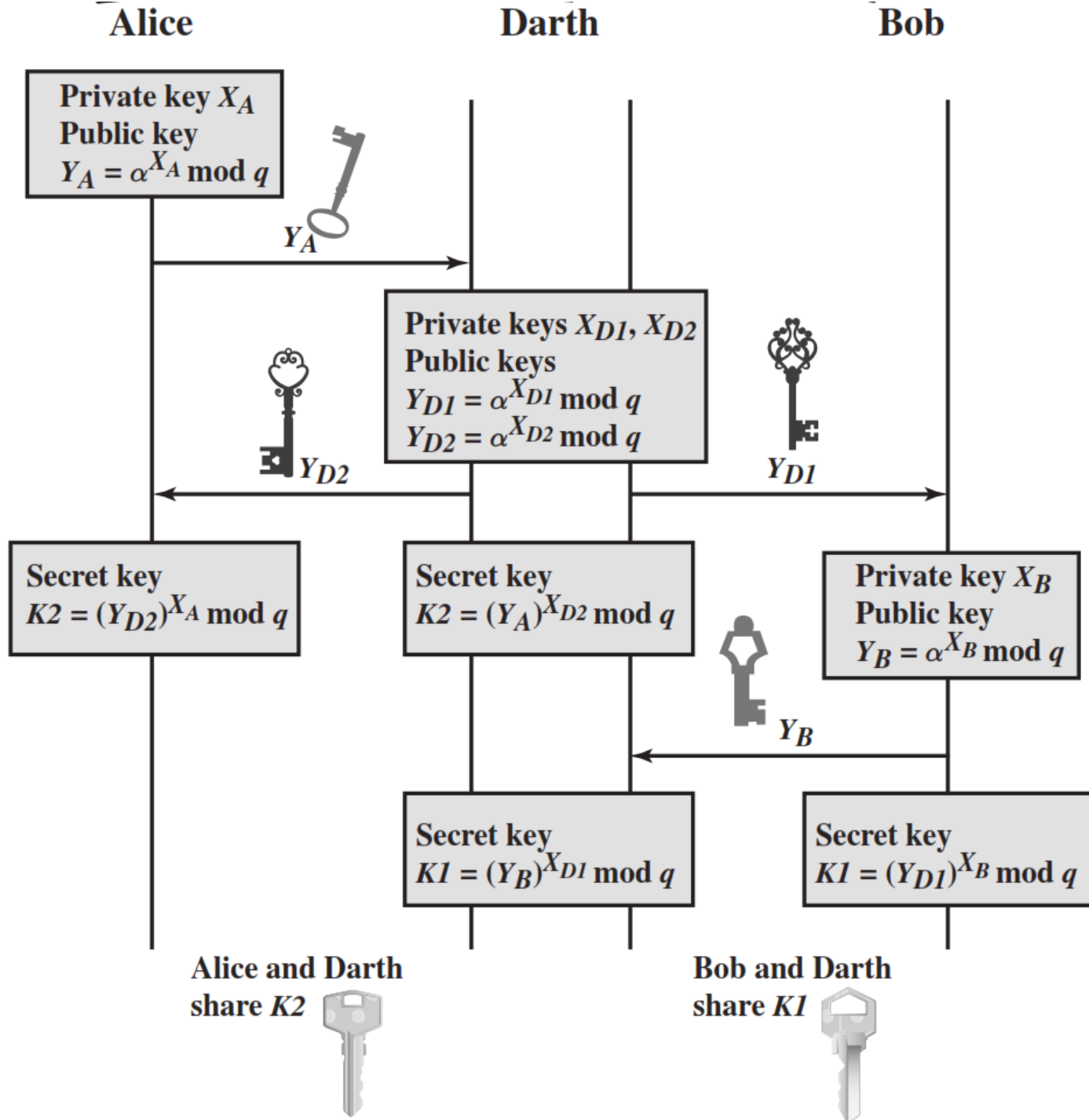
Alice calculates shared secret key $K = (Y_B)^{X_A} \bmod q$

Bob calculates shared secret key $K = (Y_A)^{X_B} \bmod q$



Man in the middle attack

- Suppose Alice and Bob wish to exchange keys, and Darth is the adversary.
1. Darth prepares for the attack by generating two random private keys X_{D1} and X_{D2} and then computes corresponding public keys Y_{D1} and Y_{D2} .
 2. Alice transmits Y_A to Bob.
 3. Darth intercepts Y_A and transmits Y_{D1} to Bob. Darth also calculates $K_2 = (Y_A)^{X_{D2}} \bmod q$.
 4. Bob receives Y_{D1} and calculates $K_1 = (Y_{D1})^{X_B} \bmod q$.
 5. Bob transmits Y_B to Alice.
 6. Darth intercepts Y_B and transmits Y_{D2} to Alice. Darth calculates $K_1 = (Y_B)^{X_{D1}} \bmod q$.
 7. Alice receives Y_{D2} and calculates $K_2 = (Y_{D2})^{X_A} \bmod q$.





**THANK
YOU FOR
LISTENING
ANY
QUESTION ?**



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